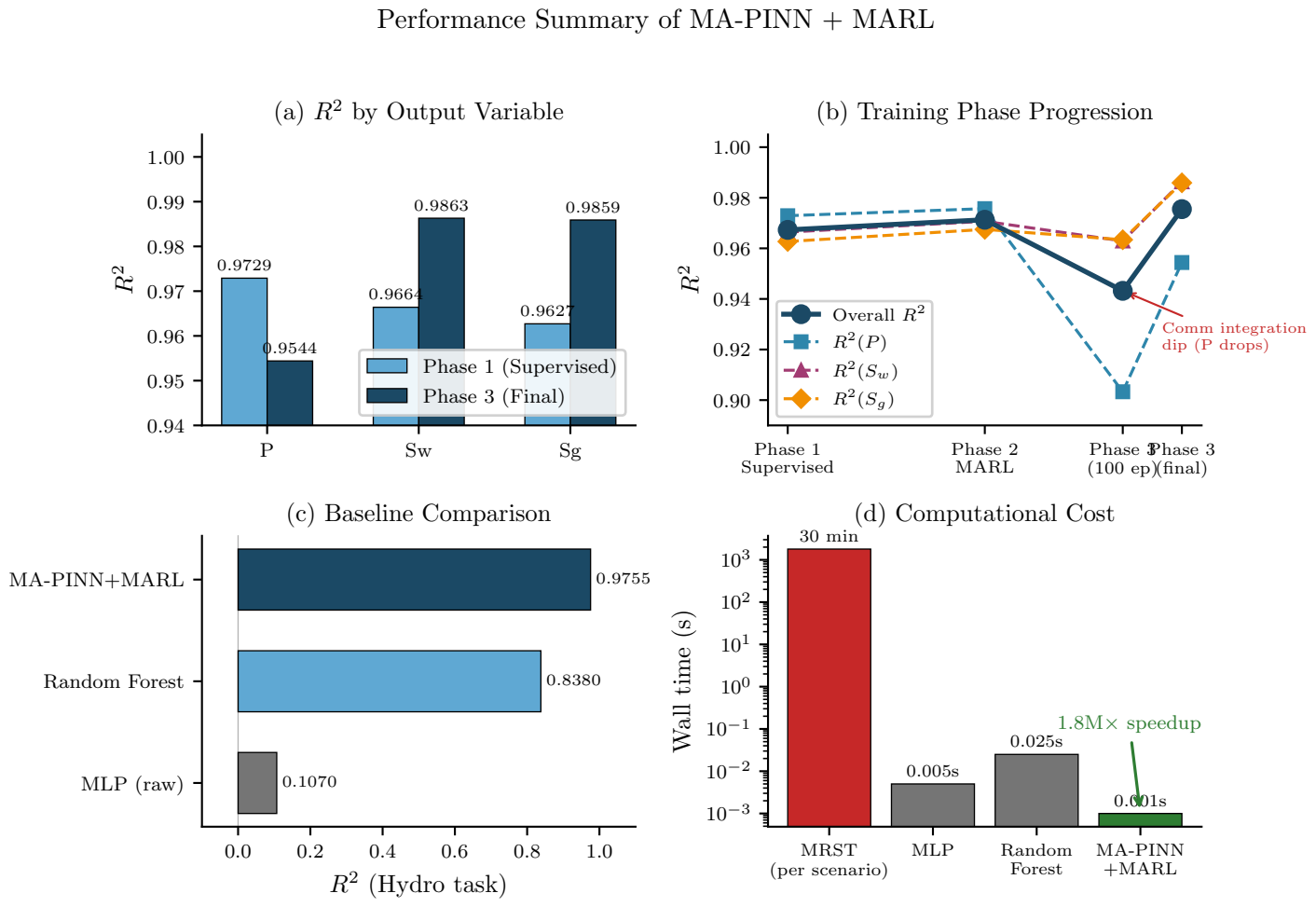
Performance Summary of MA-PINN + MARL
(a) R^2 by Output Variable
R^2
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.9729
0.9544
0.9664
0.9863
0.9627
0.9859
Phase 1 (Supervised)
Phase 3 (Final)
P
Sw
Sg
(b) Training Phase Progression
R^2
1.00
0.98
0.96
0.94
0.92
0.90
Overall R^2
$R^2(P)$
$R^2(S_w)$
$R^2(S_g)$
Comm integration dip (P drops)
Phase 1 Supervised
Phase 2 MARL
Phase 3 (100 ep)
Phase 3 (final)
(c) Baseline Comparison
MA-PINN+MARL
Random Forest
MLP (raw)
0.9755
0.8380
0.1070
0.0
0.2
0.4
0.6
0.8
1.0
R^2 (Hydro task)
(d) Computational Cost
Wall time (s)
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
30 min
0.005s
0.025s
0.001s
1.8M× speedup
MRST (per scenario)
MLP
Random Forest
MA-PINN +MARL